Week:4 Predicates and Quantifiers

CS-123: Quantitative Reasoning–I

SEECS, NUST Islamabad

Why Predicate Logic?

The Limits of Propositional Logic

A Problem

Suppose we know: “Every computer connected to the university network is functioning properly.”

- No rule of **propositional** logic lets us conclude “MATH3 is functioning properly” – even though MATH3 is one of those computers!
- Likewise, from “CS2 is under attack by an intruder,” we cannot conclude “There is a computer on the network under attack” using propositional logic alone.

The Gap

Propositional logic treats each statement as an indivisible block – it cannot see **inside** a statement to reason about the *objects* and *properties* it involves.

Enter Predicate Logic

- **Predicate logic** is more powerful: it lets us express the meaning of a wide range of mathematical and everyday statements, and reason about relationships between objects.
- Two new ingredients:
  1. **Predicates** – properties that objects can have.
  2. **Quantifiers** – “for all” and “there exists,” letting us reason about *every* object or *some* object of a certain type.

Predicates

Subject and Predicate

Breaking Down a Statement

"x is greater than 3" has two parts:

- The **subject**: the variable x.
- The **predicate**: "is greater than 3" – a **property** the subject can have.

- We write $P(x)$ for "x is greater than 3," where P denotes the predicate.
- $P(x)$ is called the value of the **propositional function** P at x.

From Propositional Function to Proposition

Key Idea

A propositional function $P(x)$ is **not** a proposition by itself – it has no truth value until x is assigned a specific value. Once x **is** assigned a value, $P(x)$ **becomes** a proposition with a definite truth value.



Example 1: A Simple Predicate

Example 1

Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Example 1: A Simple Predicate

Example 1

Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Solution: Setting $x = 4$ gives $P(4)$: "$4 > 3$" – **true**.

Example 1: A Simple Predicate

Example 1

Let $P(x)$ denote the statement "$x > 3$." What are the truth values of $P(4)$ and $P(2)$?

Solution: Setting $x = 4$ gives $P(4)$: "$4 > 3$" – **true**.

Setting $x = 2$ gives $P(2)$: "$2 > 3$" – **false**.

Example 2: A Predicate About Computers

Example 2

Let $A(x)$ denote "Computer x is under attack by an intruder." Suppose only CS2 and MATH1 are currently under attack. Find $A(\text{CS1})$, $A(\text{CS2})$, $A(\text{MATH1})$.

Solution: CS1 is **not** on the list, so $A(\text{CS1})$ is **false**.

Example 2: A Predicate About Computers

Example 2

Let $A(x)$ denote "Computer x is under attack by an intruder." Suppose only CS2 and MATH1 are currently under attack. Find $A(\text{CS1})$, $A(\text{CS2})$, $A(\text{MATH1})$.

Solution: CS1 is **not** on the list, so $A(\text{CS1})$ is **false**.

CS2 and MATH1 **are** on the list, so $A(\text{CS2})$ and $A(\text{MATH1})$ are both **true**.

Predicates with Two Variables

Extending the Idea

A statement can involve **more than one** variable. For "$x = y + 3$," we write $Q(x, y)$, where Q is the predicate. Assigning values to **both** x and y gives $Q(x, y)$ a truth value.

Example 3: A Two-Variable Predicate

Example 3

Let $Q(x, y)$ denote "$x = y + 3$." What are the truth values of $Q(1, 2)$ and $Q(3, 0)$?

Example 3: A Two-Variable Predicate

Example 3

Let $Q(x, y)$ denote "$x = y + 3$." What are the truth values of $Q(1, 2)$ and $Q(3, 0)$?

Solution: $Q(1, 2)$: "$1 = 2 + 3$" – **false**.

Example 3: A Two-Variable Predicate

Example 3

Let $Q(x, y)$ denote "$x = y + 3$." What are the truth values of $Q(1, 2)$ and $Q(3, 0)$?

Solution: $Q(1, 2)$: "$1 = 2 + 3$" – **false**.

$Q(3, 0)$: "$3 = 0 + 3$" – **true**.

Example 4: Computers and Networks

Example 4

Let $A(c, n)$ denote "Computer c is connected to network n." MATH1 is connected to CAMPUS2, but **not** to CAMPUS1. Find $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$.

Example 4: Computers and Networks

Example 4

Let $A(c, n)$ denote "Computer c is connected to network n." MATH1 is connected to CAMPUS2, but **not** to CAMPUS1. Find $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$.

Solution: $A(\text{MATH1}, \text{CAMPUS1})$ is **false** (not connected).

Example 4: Computers and Networks

Example 4

Let $A(c, n)$ denote "Computer c is connected to network n." MATH1 is connected to CAMPUS2, but **not** to CAMPUS1. Find $A(\text{MATH1}, \text{CAMPUS1})$ and $A(\text{MATH1}, \text{CAMPUS2})$.

Solution: $A(\text{MATH1}, \text{CAMPUS1})$ is **false** (not connected).

$A(\text{MATH1}, \text{CAMPUS2})$ is **true** (connected).

Three Variables, and Beyond

Going Further

We can let $R(x, y, z)$ denote "$x+y = z$." Assigning values to all three variables gives $R(x, y, z)$ a truth value.

Example 5: A Three-Variable Predicate

Example 5

What are the truth values of $R(1, 2, 3)$ and $R(0, 0, 1)$?

Example 5: A Three-Variable Predicate

Example 5

What are the truth values of $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution: $R(1, 2, 3)$: "$1 + 2 = 3$" – **true**.

Example 5: A Three-Variable Predicate

Example 5

What are the truth values of $R(1, 2, 3)$ and $R(0, 0, 1)$?

Solution: $R(1, 2, 3)$: "$1 + 2 = 3$" – **true**.

$R(0, 0, 1)$: "$0 + 0 = 1$" – **false**.

n-Place (or n-ary) Predicates

General Definition

A statement involving the n variables $x_1, x_2, \ldots, x_n$ can be denoted $P(x_1, x_2, \ldots, x_n)$. This is the value of the **propositional function** P at the n-tuple $(x_1, \ldots, x_n)$, and P is called an n-**place** (or n-**ary**) predicate.

- 1-place: $P(x)$ 2-place: $Q(x, y)$ 3-place: $R(x, y, z)$ and so on.
- Propositional functions occur constantly in **computer programs**, as our next example shows.

Example 6: A Predicate Inside a Program

Example 6

Consider the statement:

```
if x > 0 then x := x + 1
```

Example 6: A Predicate Inside a Program

Example 6

Consider the statement:

```
if x > 0 then x := x + 1
```

Explanation: The current value of x is inserted into $P(x)$: "$x > 0$."

Example 6: A Predicate Inside a Program

Example 6

Consider the statement:

```
if x > 0 then x := x + 1
```

Explanation: The current value of x is inserted into $P(x)$: "$x > 0$."

If $P(x)$ is **true**, the assignment `x:=x+1` executes – x increases by 1.

Example 6: A Predicate Inside a Program

Example 6

Consider the statement:

```
if x > 0 then x := x + 1
```

Explanation: The current value of x is inserted into $P(x)$: "$x > 0$."

If $P(x)$ is **true**, the assignment `x:=x+1` executes – x increases by 1.

If $P(x)$ is **false**, the assignment is **skipped** – x stays the same.

Preconditions and Postconditions

Verifying Program Correctness

Predicates help **prove** that a program always produces the correct output for *all* valid input – something testing alone can never fully guarantee (unless every input is tested!).

- A **precondition** describes valid input.
- A **postcondition** describes what the output should satisfy.

Example 7: Verifying a Swap Program

Example 7

```
temp := x
x := y
y := temp
```

Find a precondition and postcondition to verify this program swaps x and y.

Example 7: Verifying a Swap Program

Example 7

```
temp := x
x := y
y := temp
```

Find a precondition and postcondition to verify this program swaps x and y.

Precondition $P(x, y)$: "$x = a$ and $y = b$" (the values *before* running).
Postcondition $Q(x, y)$: "$x = b$ and $y = a$" (swapped!).

Example 7: Tracing the Proof

Assume the precondition holds: $x = a,\ y = b$.

After step	x	`temp`	y
`temp:=x`			
`x:=y`			
`y:=temp`			

Example 7: Tracing the Proof

Assume the precondition holds: $x = a,\ y = b$.

After step	x	`temp`	y
`temp:=x`	a	a	b
`x:=y`			
`y:=temp`			

Example 7: Tracing the Proof

Assume the precondition holds: $x = a,\ y = b$.

After step	x	`temp`	y
`temp:=x`	a	a	b
`x:=y`	b	a	b
`y:=temp`			

Example 7: Tracing the Proof

Assume the precondition holds: $x = a,\ y = b$.

After step	x	`temp`	y
`temp:=x`	a	a	b
`x:=y`	b	a	b
`y:=temp`	b	a	a

Example 7: Tracing the Proof

Assume the precondition holds: $x = a,\ y = b$.

After step	x	`temp`	y
`temp:=x`	a	a	b
`x:=y`	b	a	b
`y:=temp`	b	a	a

Final state: $x = b, y = a$ **– exactly the postcondition** $Q(x, y)$**! The program is verified correct.**

Practice: Evaluating Predicates

✎ Unsolved – Try These

Let $P(x)$: "$x \leq 4$." Find:

1. $P(0)$ 2) $P(4)$ 3) $P(6)$

Practice: Evaluating Predicates

✎ Unsolved – Try These

Let $P(x)$: "$x \leq 4$." Find:

1. $P(0)$ 2) $P(4)$ 3) $P(6)$

Answers: 1) True ($0 \leq 4$) 2) True ($4 \leq 4$) 3) False ($6 \not\leq 4$)

Practice: A Predicate on Words

✎ Try This

Let $P(x)$: "The word x contains the letter a." Find:

1. $P(\text{orange})$ 2) $P(\text{lemon})$

Practice: A Predicate on Words

✎ Try This

Let $P(x)$: "The word x contains the letter a." Find:

1. $P(\text{orange})$ 2) $P(\text{lemon})$

Answers: 1) True ("or**a**nge") 2) False ("lemon" has no letter a)

The Universal Quantifier

Quantification: A New Way to Build Propositions

Beyond Assigning Values

Assigning a specific value to x is **one** way to turn $P(x)$ into a proposition. **Quantification** is another: it expresses **how much** of the domain makes $P(x)$ true.

- English words: *all*, *some*, *many*, *none*, *few*.
- We focus on two: **universal** quantification (true for *every* element) and **existential** quantification (true for *some* element).
- This branch of logic is called the **predicate calculus**.

The Domain of Discourse

Critical Requirement

Every quantified statement needs a specified **domain of discourse** (or **universe of discourse**) – the set of possible values for the variable. **Without** a domain, a quantified statement has **no meaning**!

Domain Matters

The **same** predicate can be true over one domain and false over another – we'll see this vividly in Example 12.

Definition 1: The Universal Quantifier

Definition 1

The **universal quantification** of $P(x)$ is the statement "$P(x)$ for all values of x in the domain." The notation $\forall x P(x)$ denotes this. We read it "for all x, $P(x)$" or "for every x, $P(x)$."

An element for which $P(x)$ is **false** is called a **counterexample** to $\forall x P(x)$.

Example 8: A True Universal Statement

Example 8

Let $P(x)$ be "$x + 1 > x$." What is the truth value of $\forall x P(x)$, domain = all real numbers?

Example 8: A True Universal Statement

Example 8

Let $P(x)$ be "$x + 1 > x$." What is the truth value of $\forall x P(x)$, domain = all real numbers?

Solution: $P(x)$ is true for **every** real number x (adding 1 always makes it bigger). So $\forall x P(x)$ is **true**.

Remark: The Empty Domain

A Subtle Point

We usually assume domains are **nonempty**. But if the domain *were* empty, $\forall x P(x)$ would be **true** for *any* $P(x)$ – there are simply no elements to serve as counterexamples!

Other Ways to Say “For All”

Phrasings of Universal Quantification	
“for all”	“for every”
“all of”	“for each”
“given any”	“for arbitrary”
“for any”	

Careful with “Any”!

“For any *x*” is often **ambiguous** – it’s unclear whether it means “every” or “some.” It *is* unambiguous in negatives, like “There is not any reason to avoid studying.”

Counterexamples: Proving $\forall x P(x)$ False

The Key Tool

$\forall x P(x)$ is **false** if and only if $P(x)$ is not always true – i.e., there is a **counterexample**. **One** counterexample is all it takes!

Example 9: Finding a Counterexample

Example 9

Let $Q(x)$ be "$x < 2$." What is the truth value of $\forall x Q(x)$, domain = all real numbers?

Example 9: Finding a Counterexample

Example 9

Let $Q(x)$ be "$x < 2$." What is the truth value of $\forall x Q(x)$, domain = all real numbers?

Solution: $Q(3)$: "$3 < 2$" is **false** – $x = 3$ is a counterexample.

Example 9: Finding a Counterexample

Example 9

Let $Q(x)$ be "$x < 2$." What is the truth value of $\forall x Q(x)$, domain = all real numbers?

Solution: $Q(3)$: "$3 < 2$" is **false** – $x = 3$ is a counterexample.

So $\forall x Q(x)$ is **false**.

Example 10: A Counterexample with Integers

Example 10

Let $P(x)$ be "$x^2 > 0$," domain = all integers. Show $\forall x P(x)$ is false.

Example 10: A Counterexample with Integers

Example 10

Let $P(x)$ be "$x^2 > 0$," domain = all integers. Show $\forall x P(x)$ is false.

Solution: $x = 0$ gives $0^2 = 0$, which is **not** greater than 0. So $x = 0$ is a counterexample, and $\forall x P(x)$ is **false**.

Example 11: Translating a Universal Statement

Example 11

What does $\forall x N(x)$ mean if $N(x)$ is "Computer x is connected to the network," domain = all computers on campus?

Example 11: Translating a Universal Statement

Example 11

What does $\forall x N(x)$ mean if $N(x)$ is "Computer x is connected to the network," domain = all computers on campus?

Solution: "**Every computer on campus is connected to the network.**"

Example 12: The Domain Changes Everything!

Example 12

What is the truth value of $\forall x(x^2 \geq x)$, if the domain is (a) all real numbers? (b) all integers?

Example 12: The Domain Changes Everything!

Example 12

What is the truth value of $\forall x(x^2 \geq x)$, if the domain is (a) all real numbers? (b) all integers?

(a) Real numbers: False! Try $x = \frac{1}{2}$: $(\frac{1}{2})^2 = \frac{1}{4} \not\geq \frac{1}{2}$.

Example 12: The Domain Changes Everything!

Example 12

What is the truth value of $\forall x(x^2 \geq x)$, if the domain is (a) all real numbers? (b) all integers?

(a) Real numbers: False! Try $x = \frac{1}{2}$: $(\frac{1}{2})^2 = \frac{1}{4} \not\geq \frac{1}{2}$.

(b) Integers: True! The inequality $x^2 \geq x$ fails only for $0 < x < 1$ – and there are **no integers** in that range.

Why Example 12 Matters

The Big Lesson

The **exact same** predicate, $x^2 \geq x$, is **false** over the reals but **true** over the integers. **Always** state – and check – your domain before evaluating a quantified statement.

Practice: Universal Quantifier

✎ Try These

Determine the truth value, domain = all integers:

1. $\forall n(n + 1 > n)$
2. $\forall n(3n \leq 4n)$

Practice: Universal Quantifier

✎ Try These

Determine the truth value, domain = all integers:

1. $\forall n(n + 1 > n)$
2. $\forall n(3n \leq 4n)$

Answers: 1) **True** – adding 1 always increases an integer.
2) **False** – counterexample $n = -1$: $3(-1) = -3$, $4(-1) = -4$, and $-3 \not\leq -4$.

Practice: Real Numbers

✎ Try These

Determine the truth value, domain = all real numbers:

1. $\forall x((-x)^2 = x^2)$
2. $\forall x(2x > x)$

Practice: Real Numbers

✎ Try These

Determine the truth value, domain = all real numbers:

1. $\forall x((-x)^2 = x^2)$
2. $\forall x(2x > x)$

Answers: 1) **True** – squaring removes the sign, always.
2) **False** – counterexample $x = -1$: $2(-1) = -2$, and $-2 \not> -1$. (Also false for $x = 0$.)

The Existential Quantifier

Definition 2: The Existential Quantifier

Definition 2

The **existential quantification** of $P(x)$ is: "There exists an element x in the domain such that $P(x)$." Notation: $\exists x P(x)$. Here $\exists$ is the **existential quantifier**.

Phrasings	
"there exists"	"for some"
"for at least one"	"there is"

Table 1: Quantifiers Summarized

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

Key Symmetry

$\forall$ and $\exists$ are **mirror images**: one requires *every* element to work; the other requires *just one*.

Example 13: A True Existential Statement

Example 13

Let $P(x)$ be "$x > 3$." What is the truth value of $\exists x P(x)$, domain = all real numbers?

Example 13: A True Existential Statement

Example 13

Let $P(x)$ be "$x > 3$." What is the truth value of $\exists x P(x)$, domain = all real numbers?

Solution: "$x > 3$" is true for, e.g., $x = 4$. So $\exists x P(x)$ is **true**.

Example 14: A False Existential Statement

Example 14

Let $Q(x)$ be "$x = x + 1$." What is the truth value of $\exists x Q(x)$, domain = all real numbers?

Example 14: A False Existential Statement

Example 14

Let $Q(x)$ be "$x = x + 1$." What is the truth value of $\exists x Q(x)$, domain = all real numbers?

Solution: $Q(x)$ is **false** for *every* real number (no number equals itself plus 1). So $\exists x Q(x)$ is **false**.

Remark: The Empty Domain, Revisited

Existential + Empty Domain

If the domain is **empty**, $\exists x Q(x)$ is **always false** – there are no elements at all, so certainly none for which $Q(x)$ could be true. *(This is the opposite of the universal case!)*

The Uniqueness Quantifier

Uniqueness Quantifier: $\exists!$ or $\exists^1$

$\exists! x P(x)$ (or $\exists^1 x P(x)$) means "There exists a **unique** x such that $P(x)$ is true." Also phrased "there is exactly one" or "there is one and only one."

Example

$\exists! x(x - 1 = 0)$, domain = reals: states there is a **unique** real x with $x - 1 = 0$. **True** – $x = 1$ is the only solution.

A Design Choice

Uniqueness can always be **expressed** using just $\forall$ and $\exists$ (we'll see how in Section 1.5) – so it's best to stick with the two standard quantifiers when possible.

Practice: Existential Quantifier

✎ Try These

Domain = all integers. Determine the truth value:

1. $\exists n(2n = 3n)$
2. $\exists n(n = -n)$

Practice: Existential Quantifier

✎ Try These

Domain = all integers. Determine the truth value:

1. $\exists n(2n = 3n)$
2. $\exists n(n = -n)$

Answers: 1) **True** – $n = 0$: $2(0) = 0 = 3(0)$. ✓
2) **True** – $n = 0$: $0 = -0$. ✓

Practice: Real Numbers

✎ Try These

Domain = all real numbers. Determine the truth value:

1. $\exists x(x^3 = -1)$
2. $\exists x(x^2 = -1)$

Practice: Real Numbers

✎ Try These

Domain = all real numbers. Determine the truth value:

1. $\exists x(x^3 = -1)$
2. $\exists x(x^2 = -1)$

Answers: 1) **True** – $x = -1$: $(-1)^3 = -1$. ✓
2) **False** – no real number squares to a negative number.

Practice: Uniqueness

✎ Try These

Domain = all integers. Determine the truth value:

1. $\exists! x(x > 1)$
2. $\exists! x(x^2 = 1)$

Practice: Uniqueness

✎ Try These

Domain = all integers. Determine the truth value:

1. $\exists! x(x > 1)$
2. $\exists! x(x^2 = 1)$

Answers: 1) **False** – many integers satisfy $x > 1$ (2, 3, 4, …), not unique.
2) **False** – *two* integers work: $x = 1$ and $x = -1$, not unique.

Part 5

Quantifiers Over Finite Domains

Turning Quantifiers into AND/OR

The Key Insight

When the domain is **finite** – say $x_1, x_2, \ldots, x_n$ – quantified statements become ordinary propositional logic!

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \cdots \wedge P(x_n)$$

$$\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \cdots \vee P(x_n)$$

- "For all" becomes a giant **AND**.
- "There exists" becomes a giant **OR**.

Example 15: Universal over a Finite Domain

Example 15

What is the truth value of $\forall x P(x)$, where $P(x)$: "$x^2 < 10$," domain = positive integers not exceeding 4?

Example 15: Universal over a Finite Domain

Example 15

What is the truth value of $\forall x P(x)$, where $P(x)$: "$x^2 < 10$," domain = positive integers not exceeding 4?

Solution: $\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3) \wedge P(4)$.

Example 15: Universal over a Finite Domain

Example 15

What is the truth value of $\forall x P(x)$, where $P(x)$: "$x^2 < 10$," domain = positive integers not exceeding 4?

Solution: $\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3) \wedge P(4)$.

$P(4)$: "$16 < 10$" is **false** – so the whole conjunction is **false**. $\forall x P(x)$ is **false**.

Example 16: Existential over a Finite Domain

Example 16

What is the truth value of $\exists x P(x)$, where $P(x)$: "$x^2 > 10$," domain = positive integers not exceeding 4?

Example 16: Existential over a Finite Domain

Example 16

What is the truth value of $\exists x P(x)$, where $P(x)$: "$x^2 > 10$," domain = positive integers not exceeding 4?

Solution: $\exists x P(x) \equiv P(1) \vee P(2) \vee P(3) \vee P(4)$.

Example 16: Existential over a Finite Domain

Example 16

What is the truth value of $\exists x P(x)$, where $P(x)$: "$x^2 > 10$," domain = positive integers not exceeding 4?

Solution: $\exists x P(x) \equiv P(1) \vee P(2) \vee P(3) \vee P(4)$.

$P(4)$: "$16 > 10$" is **true** – so the whole disjunction is **true**. $\exists x P(x)$ is **true**.

Connecting Quantifiers to Looping



- $\forall x P(x)$: loop through all n values; if you ever find $P(x)$ false, stop – it's false. Otherwise, true.
- $\exists x P(x)$: loop searching for a true $P(x)$; find one, stop – true. Never find one, it's false.

Practice: Finite Domain Expansion

✎ Try This

Domain of $P(x) = \{0, 1, 2, 3, 4\}$. Write $\exists x P(x)$ and $\forall x \neg P(x)$ without quantifiers.

Practice: Finite Domain Expansion

✎ Try This

Domain of $P(x) = \{0, 1, 2, 3, 4\}$. Write $\exists x P(x)$ and $\forall x \neg P(x)$ without quantifiers.
Answers: $\exists x P(x) \equiv P(0) \vee P(1) \vee P(2) \vee P(3) \vee P(4)$
$\forall x \neg P(x) \equiv \neg P(0) \wedge \neg P(1) \wedge \neg P(2) \wedge \neg P(3) \wedge \neg P(4)$

Quantifiers with Restricted Domains

Restricting a Quantifier’s Range

Abbreviated Notation

We can write a condition right after the quantifier to **restrict** which elements it ranges over – without changing the whole domain.

Example 17: Restricted Quantifiers

Example 17

What do $\forall x < 0\,(x^2 > 0)$, $\forall y \neq 0\,(y^3 \neq 0)$, and $\exists z > 0\,(z^2 = 2)$ mean? (Domain: all reals in each case.)

Example 17: Restricted Quantifiers

Example 17

What do $\forall x < 0\,(x^2 > 0)$, $\forall y \neq 0\,(y^3 \neq 0)$, and $\exists z > 0\,(z^2 = 2)$ mean? (Domain: all reals in each case.)

(1) "The square of a negative real number is positive." Equivalent to $\forall x(x < 0 \rightarrow x^2 > 0)$.

Example 17: Restricted Quantifiers

Example 17

What do $\forall x < 0\,(x^2 > 0)$, $\forall y \neq 0\,(y^3 \neq 0)$, and $\exists z > 0\,(z^2 = 2)$ mean? (Domain: all reals in each case.)

(1) "The square of a negative real number is positive." Equivalent to $\forall x(x < 0 \to x^2 > 0)$.

(2) "The cube of every nonzero real number is nonzero." Equivalent to $\forall y(y \neq 0 \to y^3 \neq 0)$.

Example 17: Restricted Quantifiers

Example 17

What do $\forall x < 0\,(x^2 > 0)$, $\forall y \neq 0\,(y^3 \neq 0)$, and $\exists z > 0\,(z^2 = 2)$ mean? (Domain: all reals in each case.)

(1) "The square of a negative real number is positive." Equivalent to $\forall x(x < 0 \to x^2 > 0)$.

(2) "The cube of every nonzero real number is nonzero." Equivalent to $\forall y(y \neq 0 \to y^3 \neq 0)$.

(3) "There is a positive square root of 2." Equivalent to $\exists z(z > 0 \wedge z^2 = 2)$.

The Pattern: Restriction Rules

Two Key Patterns

Restricted universal $\rightarrow$ becomes a **conditional**:

$$\forall x{<}0\,(P(x)) \;\equiv\; \forall x(x < 0 \rightarrow P(x))$$

Restricted existential $\rightarrow$ becomes a **conjunction**:

$$\exists x{>}0\,(P(x)) \;\equiv\; \exists x(x > 0 \wedge P(x))$$

Practice: Restricted Domains

✎ Try This

Rewrite $\exists x{\geq}0\,(x^2 = 4)$ without the restricted notation, and describe it in English.

Practice: Restricted Domains

✎ Try This

Rewrite $\exists x{\geq}0\,(x^2 = 4)$ without the restricted notation, and describe it in English.
Answer: $\exists x(x \geq 0 \wedge x^2 = 4)$ – "There is a non-negative real number whose square is 4." (True: $x = 2$.)

Precedence of Quantifiers

Quantifiers Bind Tightly

Precedence Rule

The quantifiers $\forall$ and $\exists$ have **higher precedence** than **all** logical operators from propositional calculus.

Example

$\forall x P(x) \lor Q(x)$ means $(\forall x P(x)) \lor Q(x)$ – **not** $\forall x (P(x) \lor Q(x))$!

Use Parentheses!

These two readings mean **very different things**. When in doubt, always add parentheses to make your intended scope crystal clear.

Binding Variables

Bound vs. Free Variables

Definitions

When a quantifier is used on variable x, that occurrence is **bound**. A variable that is **not** bound by a quantifier (and has no assigned value) is **free**.

The Goal

Every variable in a propositional function must be **bound** (or assigned a value) to turn it into an actual proposition.

Scope of a Quantifier

Scope

The **scope** of a quantifier is the part of the expression it applies to. A variable is **free** if it's outside the scope of every quantifier that names it.

Example 18: Bound vs. Free

Example 18, Part 1

In $\exists x(x + y = 1)$: which variables are bound, which are free?

Example 18: Bound vs. Free

Example 18, Part 1

In $\exists x(x + y = 1)$: which variables are bound, which are free?

Solution: x is **bound** by $\exists x$. y is **free** – no quantifier or value binds it.

Example 18: Multiple Quantifiers, Non-Overlapping Scopes

Example 18, Part 2

In $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$: identify each quantifier's scope.

Example 18: Multiple Quantifiers, Non-Overlapping Scopes

Example 18, Part 2

In $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$: identify each quantifier's scope.

Solution: The scope of $\exists x$ is $P(x) \wedge Q(x)$ (binds x there). The scope of $\forall x$ is $R(x)$ (binds x there). **All** variables are bound – this is a full proposition once P, Q, R are specified.

Example 18: Multiple Quantifiers, Non-Overlapping Scopes

Example 18, Part 2

In $\exists x(P(x) \wedge Q(x)) \vee \forall x R(x)$: identify each quantifier's scope.

Solution: The scope of $\exists x$ is $P(x) \wedge Q(x)$ (binds x there). The scope of $\forall x$ is $R(x)$ (binds x there). **All** variables are bound – this is a full proposition once P, Q, R are specified.

Since the two scopes don't overlap, we could equally write $\exists x(P(x) \wedge Q(x)) \vee \forall y R(y)$ using a different letter – same meaning!

Practice: Identify Bound and Free Variables

✎ Try This

In $\forall x(x^2 \geq 0) \wedge (y < 5)$: which variable(s) are bound? Which are free?

Practice: Identify Bound and Free Variables

✎ Try This

In $\forall x(x^2 \geq 0) \wedge (y < 5)$: which variable(s) are bound? Which are free?
Answer: x is **bound** (by $\forall x$). y is **free** – this expression is **not yet** a full proposition until y gets a value or a quantifier.

Logical Equivalences with Quantifiers

Definition 3: Equivalence with Quantifiers

Definition 3

Statements with predicates and quantifiers are **logically equivalent** if and only if they have the **same truth value** no matter **which** predicates are substituted in and **which** domain is used. Notation: $S \equiv T$.

Example 19: Distributing $\forall$ over $\wedge$

Example 19

Show $\forall x(P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$.

Strategy: Prove Both Directions

We must show: (1) if the left side is true, the right side is true; and (2) if the right side is true, the left side is true.

Example 19: Direction 1

Suppose $\forall x(P(x) \wedge Q(x))$ is true.

Example 19: Direction 1

Suppose $\forall x(P(x) \wedge Q(x))$ is true.

Then for every element a in the domain: $P(a) \wedge Q(a)$ is true.

Example 19: Direction 1

Suppose $\forall x(P(x) \wedge Q(x))$ is true.

Then for every element a in the domain: $P(a) \wedge Q(a)$ is true.

So $P(a)$ is true **and** $Q(a)$ is true, for every a.

Example 19: Direction 1

Suppose $\forall x(P(x) \wedge Q(x))$ **is true.**

Then for every element a in the domain: $P(a) \wedge Q(a)$ is true.

So $P(a)$ is true **and** $Q(a)$ is true, for every a.

Therefore $\forall x P(x)$ is true, and $\forall x Q(x)$ is true.

Example 19: Direction 1

Suppose $\forall x(P(x) \wedge Q(x))$ is true.

Then for every element a in the domain: $P(a) \wedge Q(a)$ is true.

So $P(a)$ is true **and** $Q(a)$ is true, for every a.

Therefore $\forall x P(x)$ is true, and $\forall x Q(x)$ is true.

So $\forall x P(x) \wedge \forall x Q(x)$ is **true**. ✓

Example 19: Direction 2

Suppose $\forall x P(x) \wedge \forall x Q(x)$ is true.

Both directions hold – the equivalence is proved!

Example 19: Direction 2

Suppose $\forall x P(x) \wedge \forall x Q(x)$ is true.

Then $\forall x P(x)$ is true, and $\forall x Q(x)$ is true.

Both directions hold – the equivalence is proved!

Example 19: Direction 2

Suppose $\forall x P(x) \wedge \forall x Q(x)$ is true.

Then $\forall x P(x)$ is true, and $\forall x Q(x)$ is true.

So for every a: $P(a)$ is true, and $Q(a)$ is true.

Both directions hold – the equivalence is proved!

Example 19: Direction 2

Suppose $\forall x P(x) \wedge \forall x Q(x)$ is true.

Then $\forall x P(x)$ is true, and $\forall x Q(x)$ is true.

So for every a: $P(a)$ is true, and $Q(a)$ is true.

Hence $P(a) \wedge Q(a)$ is true for every a.

Both directions hold – the equivalence is proved!

Example 19: Direction 2

Suppose $\forall x P(x) \wedge \forall x Q(x)$ is true.

Then $\forall x P(x)$ is true, and $\forall x Q(x)$ is true.

So for every a: $P(a)$ is true, and $Q(a)$ is true.

Hence $P(a) \wedge Q(a)$ is true for every a.

So $\forall x(P(x) \wedge Q(x))$ is **true**. ✓

Both directions hold – the equivalence is proved!

What About $\vee$?

The Existential Version

By similar reasoning: $\exists x(P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$.

But NOT These!

- $\forall x(P(x) \vee Q(x)) \not\equiv \forall x P(x) \vee \forall x Q(x)$ (Exercise 52)
- $\exists x(P(x) \wedge Q(x)) \not\equiv \exists x P(x) \wedge \exists x Q(x)$ (Exercise 53)

$\forall$ distributes over $\wedge$; $\exists$ distributes over $\vee$ – but **not** the other way around!

Why $\forall$ Doesn't Distribute over $\vee$

Counterexample

Let $P(x)$: "x is even," $Q(x)$: "x is odd," domain = integers.

Why $\forall$ Doesn't Distribute over $\vee$

Counterexample

Let $P(x)$: "x is even," $Q(x)$: "x is odd," domain = integers.

$\forall x(P(x) \vee Q(x))$: every integer is even or odd – **TRUE**.

Why $\forall$ Doesn't Distribute over $\vee$

Counterexample

Let $P(x)$: "x is even," $Q(x)$: "x is odd," domain = integers.

$\forall x(P(x) \vee Q(x))$: every integer is even or odd – **TRUE**.

$\forall x P(x) \vee \forall x Q(x)$: (every integer is even) OR (every integer is odd) – **both are FALSE**, so the whole thing is **FALSE**.

Why $\forall$ Doesn't Distribute over $\vee$

Counterexample

Let $P(x)$: "x is even," $Q(x)$: "x is odd," domain = integers.

$\forall x(P(x) \vee Q(x))$: every integer is even or odd – **TRUE**.

$\forall x P(x) \vee \forall x Q(x)$: (every integer is even) OR (every integer is odd) – **both are FALSE**, so the whole thing is **FALSE**.

True $\neq$ False – confirmed NOT equivalent!

Practice: Verify $\exists$ Distributes over $\vee$

✎ Try This

Prove $\exists x(P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$ by considering both directions, similar to Example 19.

Practice: Verify $\exists$ Distributes over $\vee$

✎ Try This

Prove $\exists x(P(x) \vee Q(x)) \equiv \exists x P(x) \vee \exists x Q(x)$ by considering both directions, similar to Example 19.

Sketch: ($\Rightarrow$) If some a makes $P(a) \vee Q(a)$ true, then a makes $P(a)$ true or makes $Q(a)$ true – either way, one of $\exists x P(x)$, $\exists x Q(x)$ holds. ($\Leftarrow$) If $\exists x P(x)$ or $\exists x Q(x)$ holds, some element makes P or Q true, so it makes $P \vee Q$ true.

Negating Quantified Expressions

Motivating Example: Negating “Every Student...”

Starting Point

“Every student in your class has taken a course in calculus.”

Motivating Example: Negating “Every Student...”

Starting Point

“Every student in your class has taken a course in calculus.”

This is $\forall x P(x)$, where $P(x)$: “x has taken calculus.”

Motivating Example: Negating “Every Student...”

Starting Point

“Every student in your class has taken a course in calculus.”

This is $\forall x P(x)$, where $P(x)$: “x has taken calculus.”

Negation in English: “There is a student in your class who has **not** taken calculus.”

Motivating Example: Negating “Every Student...”

Starting Point

“Every student in your class has taken a course in calculus.”

This is $\forall x P(x)$, where $P(x)$: “x has taken calculus.”

Negation in English: “There is a student in your class who has **not** taken calculus.”

In symbols: $\exists x \neg P(x)$.

De Morgan's Law for the Universal Quantifier

Result

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

In Words

"NOT everyone has property P" means exactly "SOME ONE lacks property P."

Proving This Equivalence

$\neg\forall x P(x)$ is true $\iff$ $\forall x P(x)$ is false.

Proving This Equivalence

$\neg\forall x P(x)$ is true $\iff$ $\forall x P(x)$ is false.

$\forall x P(x)$ is false $\iff$ there is an element x in the domain for which $P(x)$ is false.

Proving This Equivalence

$\neg\forall x P(x)$ is true $\iff$ $\forall x P(x)$ is false.

$\forall x P(x)$ is false $\iff$ there is an element x in the domain for which $P(x)$ is false.

$P(x)$ is false for some x $\iff$ $\neg P(x)$ is true for some x.

Proving This Equivalence

$\neg\forall x P(x)$ is true $\iff$ $\forall x P(x)$ is false.

$\forall x P(x)$ is false $\iff$ there is an element x in the domain for which $P(x)$ is false.

$P(x)$ is false for some x $\iff$ $\neg P(x)$ is true for some x.

$\neg P(x)$ true for some x $\iff$ $\exists x \neg P(x)$ is true.

Proving This Equivalence

$\neg\forall x P(x)$ is true $\iff$ $\forall x P(x)$ is false.

$\forall x P(x)$ is false $\iff$ there is an element x in the domain for which $P(x)$ is false.

$P(x)$ is false for some x $\iff$ $\neg P(x)$ is true for some x.

$\neg P(x)$ true for some x $\iff$ $\exists x \neg P(x)$ is true.

Chaining it all together: $\neg\forall x P(x) \equiv \exists x \neg P(x)$**.** ✓

Motivating Example: Negating “Some Student...”

Starting Point

“There is a student in this class who has taken a course in calculus.”

Motivating Example: Negating “Some Student...”

Starting Point

“There is a student in this class who has taken a course in calculus.”

This is $\exists x Q(x)$.

Motivating Example: Negating “Some Student...”

Starting Point

“There is a student in this class who has taken a course in calculus.”

This is $\exists x Q(x)$.

Negation in English: “**Every** student in this class has **not** taken calculus.”

Motivating Example: Negating “Some Student...”

Starting Point

“There is a student in this class who has taken a course in calculus.”

This is $\exists x Q(x)$.

Negation in English: “**Every** student in this class has **not** taken calculus.”

In symbols: $\forall x \neg Q(x)$.

De Morgan's Law for the Existential Quantifier

Result

$$\neg \exists x Q(x) \equiv \forall x \neg Q(x)$$

In Words

"NOT even one has property Q" means exactly "EVERYONE lacks property Q."

Table 2: De Morgan's Laws for Quantifiers

Negation	**Equivalent**	**True when?**	**False when?**
$\neg\exists x P(x)$	$\forall x \neg P(x)$	Every x: $P(x)$ false.	Some x: $P(x)$ true.
$\neg\forall x P(x)$	$\exists x \neg P(x)$	Some x: $P(x)$ false.	Every x: $P(x)$ true.

Example 20: Negating Two Everyday Statements

Example 20

Find the negations of: “There is an honest politician” and “All Americans eat cheeseburgers.”

Example 20: Negating Two Everyday Statements

Example 20

Find the negations of: "There is an honest politician" and "All Americans eat cheeseburgers."

Let $H(x)$: "x is honest." Statement: $\exists x H(x)$. Negation: $\neg\exists x H(x) \equiv \forall x \neg H(x)$ – **"Every politician is dishonest."**

Example 20: The Cheeseburger Statement

Example 20, continued

Let $C(x)$: "x eats cheeseburgers." Statement: $\forall x C(x)$.

A Language Trap

"All politicians are not honest" is **ambiguous** in English – it usually means "**Not all** politicians are honest," not "every politician is dishonest." **Avoid** this phrasing; be precise instead.

Example 20: The Cheeseburger Statement

Example 20, continued

Let $C(x)$: "x eats cheeseburgers." Statement: $\forall x C(x)$.

Negation: $\neg\forall x C(x) \equiv \exists x \neg C(x)$ – **"Some American does not eat cheeseburgers."**

A Language Trap

"All politicians are not honest" is **ambiguous** in English – it usually means "**Not all** politicians are honest," not "every politician is dishonest." **Avoid** this phrasing; be precise instead.

Example 21: Negating Math Statements

Example 21

Find the negations of $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$.

Example 21: Negating Math Statements

Example 21

Find the negations of $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$.

$\neg\forall x(x^2 > x) \equiv \exists x \neg(x^2 > x) \equiv \exists x(x^2 \leq x)$.

Example 21: Negating Math Statements

Example 21

Find the negations of $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$.

$\neg\forall x(x^2 > x) \equiv \exists x \neg(x^2 > x) \equiv \exists x(x^2 \leq x)$.

$\neg\exists x(x^2 = 2) \equiv \forall x \neg(x^2 = 2) \equiv \forall x(x^2 \neq 2)$.

Example 21: Negating Math Statements

Example 21

Find the negations of $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$.

$\neg\forall x(x^2 > x) \equiv \exists x \neg(x^2 > x) \equiv \exists x(x^2 \leq x)$.

$\neg\exists x(x^2 = 2) \equiv \forall x \neg(x^2 = 2) \equiv \forall x(x^2 \neq 2)$.

The truth value of each depends on the domain – always check!

Example 22: Combining with the Implication Equivalence

Example 22

Show $\neg\forall x(P(x) \to Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$.

Example 22: Combining with the Implication Equivalence

Example 22

Show $\neg\forall x(P(x) \to Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$.

Step 1: By De Morgan's for $\forall$: $\neg\forall x(P(x) \to Q(x)) \equiv \exists x\neg(P(x) \to Q(x))$.

Example 22: Combining with the Implication Equivalence

Example 22

Show $\neg\forall x(P(x) \to Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$.

Step 1: By De Morgan's for $\forall$: $\neg\forall x(P(x) \to Q(x)) \equiv \exists x\neg(P(x) \to Q(x))$.

Step 2: Recall (Section 1.3): $\neg(P(x) \to Q(x)) \equiv P(x) \wedge \neg Q(x)$ for **every** x.

Example 22: Combining with the Implication Equivalence

Example 22

Show $\neg\forall x(P(x) \to Q(x)) \equiv \exists x(P(x) \wedge \neg Q(x))$.

Step 1: By De Morgan's for $\forall$: $\neg\forall x(P(x) \to Q(x)) \equiv \exists x\neg(P(x) \to Q(x))$.

Step 2: Recall (Section 1.3): $\neg(P(x) \to Q(x)) \equiv P(x) \wedge \neg Q(x)$ for **every** x.

Step 3: Substituting: $\exists x(P(x) \wedge \neg Q(x))$. **QED!**

Practice: Negate These

✎ Try These

Express, then negate (no negation left of a quantifier):

1. “All dogs have fleas.”
2. “There is a horse that can add.”

Practice: Negate These

✎ Try These

Express, then negate (no negation left of a quantifier):

1. "All dogs have fleas."
2. "There is a horse that can add."

Answers: 1) $\forall x D(x)$; negation $\exists x \neg D(x)$ – "Some dog does not have fleas."
2) $\exists x H(x)$; negation $\forall x \neg H(x)$ – "No horse can add."

Practice: Counterexamples

✎ Try These

Find a counterexample, domain = all integers:

1. $\forall x(x^2 \geq x)$
2. $\forall x(x = 1)$

Practice: Counterexamples

✎ Try These

Find a counterexample, domain = all integers:

1. $\forall x(x^2 \geq x)$
2. $\forall x(x = 1)$

Answers: 1) No counterexample among integers – **true**! (Verified in Example 12.)
2) Counterexample: $x = 0$ (or any integer other than 1) – **false**.

Six Translation Patterns Worth Memorizing

English	**Pattern**
"No one is P"	$\neg\exists x P(x)$ or $\forall x \neg P(x)$
"Not everyone is P"	$\neg\forall x P(x)$ or $\exists x \neg P(x)$
"All A's are B"	$\forall x(A(x) \rightarrow B(x))$
"At least one A is B"	$\exists x(A(x) \wedge B(x))$
"Everyone is A and B"	$\forall x(A(x) \wedge B(x))$
"Not everybody is A, or someone is not B"	$\neg\forall x A(x) \vee \exists x \neg B(x)$

Watch the Connective!

"All A's are B" uses $\rightarrow$, **not** $\wedge$ – this trips up almost everyone at first!

Quantifiers in System Specifications

From Propositions to Predicates in Specs

Why It Matters

In Section 1.2 we wrote system specifications using plain propositions. But real specs talk about **every** message, **some** link, **all** users – which needs predicates and quantifiers.

Example 25: Mail Messages

Example 25, Part 1

Express: “Every mail message larger than one megabyte will be compressed.”

Example 25: Mail Messages

Example 25, Part 1

Express: “Every mail message larger than one megabyte will be compressed.”

Let $S(m, y)$: “Mail message m is larger than y megabytes” and $C(m)$: “Mail message m will be compressed.”

Example 25: Mail Messages

Example 25, Part 1

Express: “Every mail message larger than one megabyte will be compressed.”

Let $S(m, y)$: “Mail message m is larger than y megabytes” and $C(m)$: “Mail message m will be compressed.”

Translation: $\forall m\,(S(m, 1) \rightarrow C(m))$.

Example 25: Network Links

Example 25, Part 2

Express: “If a user is active, at least one network link will be available.”

Mind the Precedence!

Since quantifiers bind tighter than $\rightarrow$, this reads as $(\exists u A(u)) \rightarrow (\exists n S(n, \text{available}))$ – exactly what we intended here.

Example 25: Network Links

Example 25, Part 2

Express: “If a user is active, at least one network link will be available.”

Let $A(u)$: “User u is active,” and $S(n, x)$: “Network link n is in state x.”

Mind the Precedence!

Since quantifiers bind tighter than $\rightarrow$, this reads as $(\exists u A(u)) \rightarrow (\exists n S(n, \text{available}))$ – exactly what we intended here.

Example 25: Network Links

Example 25, Part 2

Express: “If a user is active, at least one network link will be available.”

Let $A(u)$: “User u is active,” and $S(n, x)$: “Network link n is in state x.”

Translation: $\exists u A(u) \rightarrow \exists n\, S(n, \text{available})$.

Mind the Precedence!

Since quantifiers bind tighter than $\rightarrow$, this reads as $(\exists u A(u)) \rightarrow (\exists n S(n, \text{available}))$ – exactly what we intended here.

Practice: System Specification Translation

✎ Try These

Let $S(x, y)$: "system x is in state y." Translate to English:

1. $\exists x\, S(x, \text{open})$
2. $\forall x\, (S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$

Practice: System Specification Translation

✎ Try These

Let $S(x, y)$: "system x is in state y." Translate to English:

1. $\exists x\, S(x, \text{open})$
2. $\forall x\, (S(x, \text{malfunctioning}) \vee S(x, \text{diagnostic}))$

Answers: 1) "Some system is open."
2) "Every system is either malfunctioning or in diagnostic mode."

Practice: More Specification Translation

✎ Try These

1. $\exists x \neg S(x, \text{available})$
2. $\forall x \neg S(x, \text{working})$

Practice: More Specification Translation

✎ Try These

1. $\exists x \neg S(x, \text{available})$
2. $\forall x \neg S(x, \text{working})$

Answers: 1) “Some system is not available.”
2) “No system is working.”